

The problem is framed as an inverse problem. The backward map takes position and orientation data and calculated 8 flux values from each of the coils. Our goal is to approximate an inverse to this map. To do so, we first try to understand the backward map.

First we define a position/orientation vector $\mathbf{p} \in \mathbb{R}^5$

$$\mathbf{p} = \begin{bmatrix} x \\ y \\ z \\ \theta \\ \phi \end{bmatrix} \quad (1)$$

Then we define a vector of fluxes, $\Phi \in \mathbb{R}^8$

$$\Phi = \begin{bmatrix} \Phi_1 \\ \Phi_2 \\ \Phi_3 \\ \Phi_4 \\ \Phi_5 \\ \Phi_6 \\ \Phi_7 \\ \Phi_8 \end{bmatrix} \quad (2)$$

Now we can write the backward map: $\Phi(\mathbf{p}) : \mathbb{R}^5 \rightarrow \mathbb{R}^8$. More precisely,

$$\Phi_i = \mathbf{H}_i(x, y, z) \cdot \hat{n}(\theta, \phi) \quad (3)$$

where $\mathbf{H}_i$ is the magnetic field calculated at (x, y, z) due to coil i using equation (??), and $\hat{n}(\theta, \phi)$ is the vector normal to the sensor.